



**CSE370: Database Management System  
Assignment on Functional Requirements**

**Proposed Project Title:**

**“Things We Show and Tell” - A Web-based Roguelike Card-strategy  
Game**

| Group No: 01, CSE370 Lab Section: 11,<br>Summer 2026 |          |                     |
|------------------------------------------------------|----------|---------------------|
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## Project Overview

The following is an endless roguelike strategy based card game designed around turn-based tactical combat, deck management, and continuous player tracking. The cards are classified into monsters, which are used for combat and spells, which are used to create dynamic events and change stats and gameplay. Players start session-based runs, earn gold by combat, purchase cards from a list of randomized cards for better combat, and upgrade their active deck at the merchant. The game also manages current session stats, enforces correct state for real time combat grids across 8 slots and also records high score achievements across player accounts.

## Tech Stack:

- Language: JavaScript
- Frontend Framework: React.js
- Backend Framework: Node.js with Express.js
- Styling: TailwindCSS
- Database: MySQL / SQLite / AlaSQL
- Deployment: Github Pages / Netlify

## Functional Requirements

| Member 1 |                                                                                                                                                                                                                                                                                                                                                                                                                 |
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| No.      | Feature Description                                                                                                                                                                                                                                                                                                                                                                                             |
| 1        | Player Account & High Score Tracking: Manages user accounts and records high scores for individual runs. When a new user registers, their credentials are saved as a new entry in the “Players” entity. During an active run, the game continuously tracks the score. Once it ends, it compares the high score with previous data and updates the “High Score” if a new score is achieved                       |
| 2        | Card Template Registry: The storage for all static cards in the game so that we can prevent redundancy. All cards will be stored generally along with their baseline attributes as a “Card” entity. Moreover, using EER, we can distinguish monster cards and spell cards with their specific traits. Then we can manage player inventory of owned cards and upgrades and fetch them from this place as needed. |

| Member 2 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
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| No.      | Feature Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 3        | Game Session Tracking: Manages active play sessions as players progress through the endless matches. Creating a new run creates a new record in the “Matches” table with baseline parameters like, a 100 gold, 0 score, Level 1 and a boolean value to track game state. All subsequent gameplay afterwards including combat, upgrades, purchases, inventory, are linked to the session until player health drops to zero which triggers the ending of a run and sets the required parameters accordingly and also changes the boolean to false. |
| 4        | Inventory Tracking: Tracks specific cards players get and upgrades during a run. Rather than keeping multiple copies of the same card, we use instances to keep track of changes dynamically. This record tracks modifications like level, attack etc while being linked to the “Card” catalog to dynamically resolve baseline attributes.                                                                                                                                                                                                       |

| Member 3 |                                                                                                                                                                                                                                                                                                                                                                                                                     |
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| No       | Feature Description                                                                                                                                                                                                                                                                                                                                                                                                 |
| 5        | Merchant Shop and Transaction: Handles dynamic markets generated between certain stages. The backend populated a temporary table with randomly selected cards, new and old ones which are now upgraded and are linked with the current run. Upon purchase, the database is updated and baseline stats are adjusted as required along with duplicate purchase avoidance and inserting cards in the player ownership. |

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| 6 | Combat and Slot Tracking: Maintains battlefield layout and real-time card stats during active battles. The battlefield is a grid of 8 separate slots (4 for each side with 2 monster zones and 2 spell zones). Dynamic attributes are actively tracked and it is also ensured that all effects and damage during mid-game are retained maintaining database integrity. |
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